

Advanced Machine Learning Techniques for Solving Nonlinear Mathematical Optimization Problems

Abstract

This study presents an advanced framework for solving nonlinear mathematical optimization problems using machine learning and artificial intelligence techniques. Traditional optimization methods often face challenges when dealing with high-dimensional, complex, and non-convex mathematical functions. To address these limitations, the proposed research integrates deep learning models with metaheuristic optimization algorithms to improve solution accuracy and computational efficiency. The framework employs neural-network-based prediction mechanisms to approximate optimal solutions while utilizing evolutionary search strategies for global exploration. Experimental evaluations were conducted on benchmark optimization datasets and mathematical test functions. The results demonstrate that the hybrid approach achieves superior convergence speed, reduced computational complexity, and enhanced optimization performance compared with conventional methods. The findings highlight the potential of combining machine learning with mathematical optimization to address real-world challenges in engineering, finance, transportation, and scientific computing. The proposed methodology provides a scalable and adaptive solution for next-generation intelligent optimization systems.

Keywords: Mathematical Optimization, Machine Learning, Deep Learning, Artificial Intelligence, Metaheuristic Algorithms, Nonlinear Optimization, Neural Networks, Evolutionary Computing, Computational Mathematics, Intelligent Decision-Making, Scientific Computing, Optimization Algorithms